DentTime

Data Pipeline for dental ML

ระบบเตรียมข้อมูลสำหรับโมเดลทำนายเวลาหัตถการ

1. Anonymization

HMAC + k-anonymity

→ PDPA safe

2. Airflow Pipeline

4 tasks, monthly

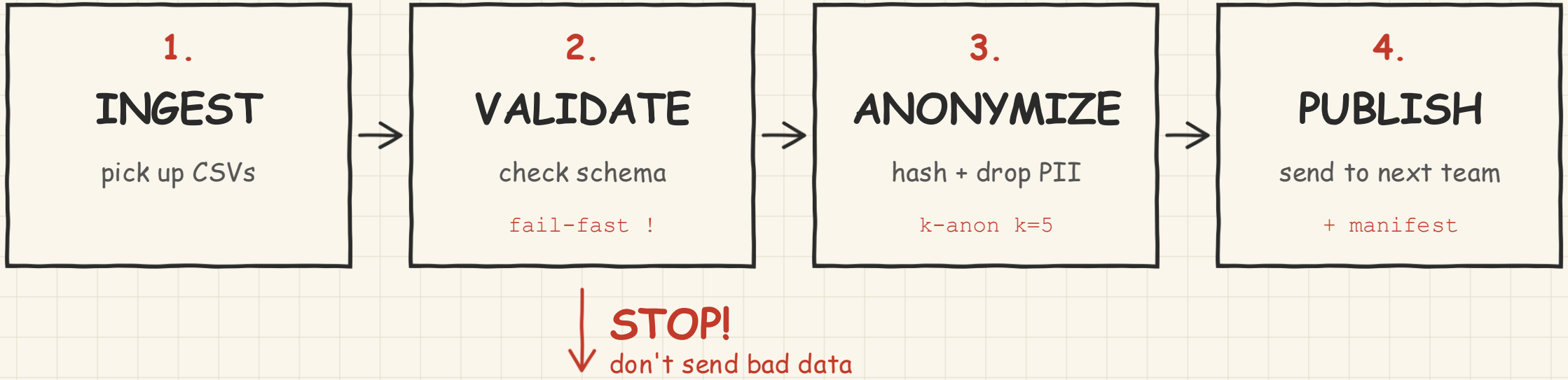
→ auto, retryable

3. Handoff

clean dataset + docs

→ 252K rows ready

The Pipeline



on retry: each task tries 2x before giving up.

idempotent — safe to re-run the same month.

logs: row counts, duration, status — every run.

Where the data goes

Start with 424K raw rows. Keep 252K clean ones.

raw CSV *3 files from DentCloud* 424,053 rows

100%

after dedupe *appointment_id duplicated* 261,407 rows

61.6%

after k-anon k=5 *drop tiny QI groups* 252,359 rows

59.5%

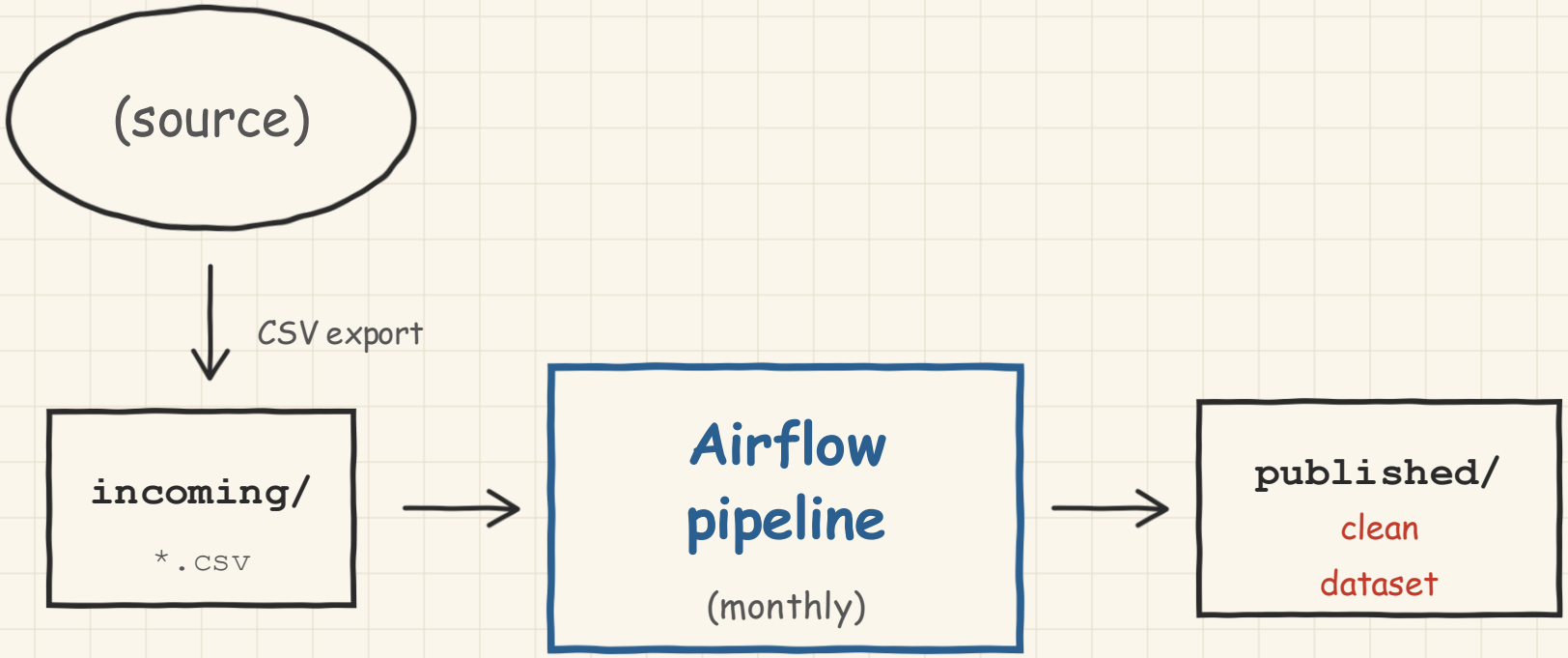
k-anon dropped only ~3.5% — most rows survive.

the big drop:

dedupe removes ~163K duplicate appointment rows.

merging 3 source CSVs caused overlap.

Why file-drop (not DB)?



WHY?

safer.
decoupled.
backfill =
drop a file.

no DB credentials in the pipeline = nothing to break in production.

What downstream gets

1 dataset · 252,359 rows · 21 columns in 5 groups

1. IDs

clinic
dentist
appointment
has_dentist_id

2. Clinical

treatment
tooth_no
surfaces,
total_amount,
has_notes

3. Time

year_month,
day_of_week,
hour_bucket

4. Behavioral

duration,
delays,
offsets,
status flags

5. Appt. Order

appointment_
rank_in_day,
is_first_case

→ target: **scheduled_duration_min**

→ split by: **appt_year_month** *(time-based)*

Things to know

tradeoffs & limits — before you use this dataset

- ☒ **IDs rotate every run**
  can't join across dataset versions
- ☒ **notes are gone**
  only has_notes flag (0/1) kept — PDPA
- ☒ **hour is fuzzy**
  4-hour bucket, not exact time (k-anon cost)
- ☒ **~50% rows missing dentist license**
  kept with has_dentist_id flag — FE uses Two-Tier Fallback